

# Hallucination Type Matters: A Taxonomy-Controlled Ablation Study on LLM-Augmented Molecular Property Prediction

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## Abstract

**Background:** This work addresses these gaps through a controlled perturbation framework to characterize *architecture-dependent sensitivity* to structured molecular fabrications. We make the following scoping explicit: we do not propose a novel architecture for molecular prediction, nor do we claim to mechanistically explain why hallucinations alter LLM reasoning. Instead, we empirically characterize how model capacity and task domain interact with specific types of structured semantic perturbations.

The primary contribution of this work is the characterization of architecture-dependent hallucination sensitivity through a controlled semantic perturbation framework. We define a lightweight categorization scheme—Structural Phantom (SP), Property Inversion (PI), Mechanism Fabrication (MF), and Contextual Confabulation (CC)—utilized as controlled perturbation operators to isolate specific semantic effects across nine conditions and two model scales (Llama-3.1-8B and Llama-3.3-70B).

**Results:** We identify a task-dependent divergence and evidence of architecture-dependent sensitivity: on BBBP, Contextual Confabulation (CC) is associated with performance shifts for the 8B model, while the 70B model exhibits reversed sensitivity. On BACE, all scales exhibit consistent distributional compression toward chance level. A random-permutation control confirms that these effects are semantically grounded rather than driven by stylistic priming.

**Conclusions:** These findings suggest that LLM-based molecular reasoning is sensitive to structured semantic perturbations in an architecture-dependent manner, highlighting the importance of scale-aware evaluation when using generative augmentations in scientific pipelines.

## 1 Introduction

Molecular property prediction is a critical component of early-stage drug discovery, enabling the prioritization of candidate compounds before expensive synthesis and biological assays [15]. Machine learning approaches—including graph neural networks [3], fingerprint-based classifiers [12], and descriptor-based models—have become standard tools, with MoleculeNet [17] providing widely adopted benchmarks across physicochemical, enzymatic, and toxicity tasks.

More recently, large language models (LLMs) have been applied to molecular reasoning by converting SMILES notation into natural language contexts for text-based inference [4, 13]. Within this line of work, an observation has been reported in prior work: hallucinated molecular descriptions—scientifically styled but factually incorrect text—have been associated with improved downstream prediction performance in specific settings [5]. This finding is in tension with the common assumption that strict factual correctness is a prerequisite for useful textual augmentation.

However, existing investigations of this phenomenon share several limitations. First, hallucination is typically treated as a monolithic category, obscuring potentially important differences among types of

factual error. Second, prior studies often lack semantic controls capable of disentangling the contribution of molecular-relevant content from generic scientific language effects (stylistic priming). Third, it remains unclear whether these hallucination-associated performance variations generalize across prediction tasks with different underlying knowledge requirements. Understanding this sensitivity is important for interpreting LLM-augmented predictions, particularly in settings where textual augmentation may introduce uncontrolled semantic perturbations.

This work addresses these gaps through a controlled perturbation framework. We make the following scoping explicit: we do not propose a novel architecture for molecular prediction, nor do we claim to mechanistically explain why hallucinations alter LLM reasoning. Instead, we empirically characterize *which types* of structured semantic perturbations are associated with performance shifts, and under *which task conditions* these associations emerge.

The primary contribution of this work is the introduction of a controlled semantic perturbation framework to systematically evaluate how structured hallucination types influence LLM-based molecular prediction. We define a lightweight, operational categorization scheme—comprising Structural Phantom (SP), Property Inversion (PI), Mechanism Fabrication (MF), and Contextual Confabulation (CC) types—utilized as controlled perturbation operators to isolate specific semantic effects. We evaluate this framework through a nine-condition ablation on two MoleculeNet benchmarks representing distinct regimes: BBBP (physicochemical) and BACE (enzyme inhibition).

Our key contributions include:

- A controlled perturbation framework enabling the systematic isolation of hallucination-associated performance variations through semantic and stylistic disruption.
- A lightweight categorization scheme for molecular description errors, operationally defined as perturbation operators to maintain interpretability and reproducibility.
- Evidence of a task-dependent divergence in hallucination sensitivity, where structurally grounded (SP) perturbations are associated with directional improvements in physicochemical prediction, while similar augmentations induce performance degradation in enzymatic tasks.
- Characterization of prediction distribution compression on BACE, where hallucination augmentation shifts bimodal baseline distributions toward non-discriminative narrow peaks.
- A random-permutation control experiment demonstrating that semantic content, rather than stylistic register alone, is the primary driver of observed performance shifts within the evaluated setting.
- Evidence of architecture-dependent hallucination sensitivity, where smaller models exhibit performance gains under free hallucination augmentation on physicochemical tasks, while larger models show reversed sensitivity, suggesting a non-linear relationship between model capacity and perturbation response.

We explicitly bound this exploratory pilot study ( $N = 204$  for BBBP,  $N = 152$  for BACE) to two LLM scales. These findings are intended to structure future empirical investigation into LLM augmentation rather than to establish definitive causal mechanisms.

## 2 Related Work

### 2.1 Molecular Property Prediction

Molecular property prediction has been standardized through benchmark suites such as MoleculeNet [17], which provides curated classification and regression tasks spanning physicochemical, biophysical, and physiological domains. State-of-the-art approaches include message-passing neural networks

(MPNNs) [3], graph transformer architectures [18], and descriptor-based models using extended-connectivity fingerprints [12]. Scaffold splitting [1] is widely adopted to evaluate generalization under structural novelty. **Distinction:** Our work is not a model development study. We do not propose a new prediction architecture. Rather, we use an existing LLM in a zero-shot setting to investigate how different types of textual augmentation affect prediction behavior.

## 2.2 LLMs in Molecular Science

Large language models have been increasingly applied to chemical tasks, including reaction prediction [14], retrosynthesis [9], and molecular property estimation [4, 13]. Text-based molecular representations, where SMILES strings are accompanied by natural language descriptions, have shown promise for zero-shot and few-shot inference [2]. Recent studies have explored whether LLMs encode implicit chemical knowledge through their pre-training corpora [16]. **Distinction:** Our work is not a prompt engineering study. We do not optimize prompts for maximum prediction accuracy. Instead, we use a fixed prompt template to systematically vary the semantic content of augmentation text while controlling for confounding factors.

## 2.3 Hallucination in Scientific LLM Applications

LLM hallucination—the generation of fluent but factually unsupported text—has been extensively studied in general NLP contexts [6, 7]. In molecular and scientific applications, hallucination poses particular risks because factual errors in structural or mechanistic claims may not be easily detected by non-expert users [10]. Recent work by Hao et al. [5] reported the counterintuitive finding that hallucinated descriptions can improve molecular property prediction, though without characterizing which aspects of the hallucinated text drive this effect. Yuan et al. [19] proposed a taxonomy for molecular hallucinations but did not conduct controlled ablation experiments isolating the contribution of each category. **Distinction:** Our work is not a hallucination detection or mitigation study. We treat hallucinations as structured semantic perturbations and study their differential effects on prediction behavior through controlled ablation, incorporating semantic and stylistic controls absent from prior evaluations.

# 3 Methods

## 3.1 Operational Categorization Scheme

We utilize an operational categorization scheme for molecular hallucinations, each type detectable through rule-based classification using RDKit [8] as the ground-truth engine. We emphasize that this scheme is a heuristic tool for structured perturbation analysis, not an exhaustive semantic ontology. The keyword sets were intentionally constrained to maintain interpretability and reproducibility rather than maximize coverage. The categories are mutually exclusive and applied in priority order (SP > PI > MF > CC):

- **Structural Phantom (SP):** The generated text references chemical elements or functional groups that are absent from the molecular structure. Detection: for a predefined heuristic dictionary of element keywords (e.g., “fluorine”, “fluoro”), we check whether the corresponding element is present in the molecule’s atom set via `mol.GetAtoms()`. If a keyword matches but the element is absent, the description is classified as SP.
- **Property Inversion (PI):** The generated text claims physicochemical properties that contradict computed descriptors. Detection: we compute MolLogP via RDKit. If  $\text{LogP} > 3$  and the text contains keywords from a predefined hydrophilic dictionary, or  $\text{LogP} < 1$  and the text contains keywords from a lipophilic dictionary, the description is classified as PI.

- **Mechanism Fabrication (MF):** The generated text names specific biological targets or pathways. Detection: exact keyword matching against a fixed list of 9 common targets (e.g., ACE2, HER2, EGFR).
- **Contextual Confabulation (CC):** The residual category. Descriptions that do not trigger SP, PI, or MF criteria are classified as CC. This category represents scientifically plausible but generic narrative hallucination.

We acknowledge that CC functions as a residual class and encompasses heterogeneous subtypes that future work should further differentiate.

## 3.2 Experimental Conditions

Our ablation framework consists of nine conditions, designed to isolate the contributions of semantic content, scientific register, and hallucination type:

- **C0 (Baseline):** SMILES notation only, with no augmentation.
- **C1 (Factual):** SMILES augmented with a factual description generated by RDKit (molecular weight, LogP, hydrogen bond donors/acceptors, TPSA, ring count, atom composition).
- **C2 (Chemical Priming):** SMILES augmented with a template-generated text using scientific-sounding but semantically vacuous language that includes chemical vocabulary. We note that C2 is *not* a pure stylistic control because it contains domain-relevant vocabulary; C2b serves as the stricter control.
- **C2b (Pure Gibberish):** SMILES augmented with template-generated text using exclusively non-chemical vocabulary. This condition strictly controls for prompt length and linguistic complexity without domain priming.
- **C3 (Free Hallucination):** SMILES augmented with an unconstrained LLM-generated description, replicating the protocol of prior work [5].
- **C4a (Structural Phantom):** SMILES augmented with a hallucination generated via constrained prompting to focus on structural features and functional groups.
- **C4b (Property Inversion):** SMILES augmented with a hallucination generated via constrained prompting to focus on solubility, lipophilicity, and permeability.
- **C4c (Mechanism Fabrication):** SMILES augmented with a hallucination generated via constrained prompting to hypothesize biological targets and cellular mechanisms.
- **C5 (Random-Permutation):** SMILES augmented with a hallucination originally generated for a *different* molecule, assigned via a fixed random permutation ( $seed = 42$ ). This preserves the scientific register and statistical properties of hallucinated text while completely disrupting semantic alignment with the target molecule.

## 3.3 Prediction Protocol

All prediction queries use a fixed zero-shot prompt template:

```
Given this molecule (SMILES: {smiles})
[Additional information: {augmentation}]
On a scale of 0 to 1, what is the probability that this molecule can {task}?
Respond with ONLY a number between 0 and 1. Nothing else.
```

The augmentation line is omitted for C0. The task description is “penetrate the blood-brain barrier” for BBBP and “inhibit the BACE-1 enzyme” for BACE. The numerical output was deterministically extracted, with malformed responses defaulting to 0.5.

### 3.4 Implementation Details

**Model.** Primary ablation experiments were conducted with **Llama-3.1-8B-Instant** via the Groq API. Cross-model validation was extended to **Llama-3.3-70B-Versatile**, as described in Section 3.5.

**Generation parameters.** Prediction queries: temperature = 0.0, max\_tokens = 10, seed = 42. Hallucination generation: temperature = 0.9, max\_tokens = 150, no fixed seed. We explicitly distinguish between stochastic hallucination generation (no fixed seed) and quasi-deterministic prediction inference (seed=42), noting that API-level non-determinism may still introduce minor variability. To ensure exact reproducibility despite stochastic generation, the static dataset containing all generated augmentations used for evaluation is provided in the repository. Failed API calls (rate limits, parsing errors) defaulted to  $p = 0.5$ ; the proportion of such defaults is reported in Appendix A.

**Data.** Datasets were loaded via DeepChem [11]. Both BBBP ( $N = 204$ ) and BACE ( $N = 152$ ) were partitioned using scaffold splitting [1] via DeepChem’s built-in splitter (80/10/10 train/valid/test). Only the held-out test set was used for evaluation.

**Factual descriptions.** For condition C1, factual descriptions were generated using RDKit molecular descriptors: molecular weight, Wildman-Crippen LogP, number of hydrogen bond donors and acceptors, topological polar surface area, heavy atom count, and ring count.

**Traditional ML Baseline.** To provide context for the zero-shot LLM predictions, we trained a Random Forest classifier (500 trees) using Morgan fingerprints (radius=2, 2048 bits) on the training and validation partitions of the scaffold split. The baseline was evaluated on the exact same test sets ( $N = 204$  for BBBP,  $N = 152$  for BACE) to ensure strict comparability.

**Evaluation.** Discriminative performance was assessed using ROC-AUC. Confidence intervals (95%) were computed via bootstrap resampling (1,000 iterations, seed = 42). To assess statistical significance of performance differences relative to baseline (C0), we used a bootstrap paired AUC difference test. Multiple comparisons were corrected using the Benjamini-Hochberg procedure.

### 3.5 Robustness and Cross-Model Validation

To ensure the stability of our findings, we performed two additional validation steps:

1. **Multi-Seed Evaluation:** For the primary Llama-3.1-8B model, we repeated the prediction protocol for conditions C0 and C3 across three independent runs using different inference seeds (42, 123, 777) to calculate mean performance and standard deviation.
2. **Cross-Model Generalization:** To verify if the observed phenomena generalize across model scales, we executed the prediction protocol for C0 and C3 on the larger **Llama-3.3-70B-Versatile** model.

This multi-layered approach was designed to distinguish systematic semantic effects from stochastic artifacts or model-specific quirks.

## 4 Results

### 4.1 Predictive Performance Across Conditions

Table 1 and Figure 1 summarize the primary ROC-AUC scores, bootstrap 95% confidence intervals, and statistical comparisons for all nine conditions on both datasets, using a fixed inference seed (42) as the primary benchmark. To assess the stability of these observations across stochastic variations, multi-seed robustness results for selected conditions are provided in Table 2.

#### 4.1.1 BBBP (Physicochemical Task)

To contextualize the LLM zero-shot performance, the traditional ML baseline (Random Forest with Morgan fingerprints) achieved an expectedly strong ROC-AUC of 0.705 on the BBBP test set. While

Table 1: Predictive performance (ROC-AUC) across nine augmentation conditions on BBBP and BACE. Bootstrap 95% confidence intervals computed over 1,000 resamples. Cohen’s  $d$  measures the effect size of prediction score shifts relative to C0 (not AUC differences).  $p$ -values are corrected for multiple comparisons using the Benjamini-Hochberg procedure. Boldface indicates the highest ROC-AUC achieved among zero-shot LLM conditions per dataset.

| Condition                      | BBBP ( $N = 204$ ) |                |                 | BACE ( $N = 152$ ) |                |                 |
|--------------------------------|--------------------|----------------|-----------------|--------------------|----------------|-----------------|
|                                | ROC-AUC            | 95% CI         | $p_{\text{BH}}$ | ROC-AUC            | 95% CI         | $p_{\text{BH}}$ |
| <i>Traditional ML Baseline</i> |                    |                |                 |                    |                |                 |
| RF (Morgan FP)                 | 0.705              | —              | —               | 0.893              | —              | —               |
| <i>LLM Zero-Shot</i>           |                    |                |                 |                    |                |                 |
| C0 (Baseline)                  | 0.533              | [0.471, 0.593] | 1.000           | 0.674              | [0.592, 0.759] | 1.000           |
| C1 (Factual)                   | 0.541              | [0.480, 0.603] | 0.846           | 0.486              | [0.402, 0.569] | 0.011           |
| C2 (Chem. Priming)             | 0.613              | [0.545, 0.685] | 0.344           | 0.522              | [0.454, 0.589] | 0.011           |
| C2b (Pure Gibberish)           | 0.553              | [0.485, 0.626] | 0.743           | 0.623              | [0.542, 0.701] | 0.217           |
| <i>Hallucination</i>           |                    |                |                 |                    |                |                 |
| C3 (Free Hallu)                | 0.593              | [0.519, 0.664] | 0.438           | 0.504              | [0.420, 0.592] | 0.016           |
| C4a (SP)                       | <b>0.626</b>       | [0.548, 0.698] | 0.344           | <b>0.666</b>       | [0.579, 0.748] | 0.808           |
| C4b (PI)                       | 0.563              | [0.480, 0.634] | 0.743           | 0.542              | [0.448, 0.628] | 0.011           |
| C4c (MF)                       | 0.611              | [0.537, 0.685] | 0.352           | 0.566              | [0.493, 0.645] | 0.083           |
| <i>Control</i>                 |                    |                |                 |                    |                |                 |
| C5 (Random-Permutation)        | 0.481              | [0.408, 0.554] | 0.438           | 0.492              | [0.405, 0.584] | 0.012           |

Note: Table 1 reports results for the primary single-run evaluation (seed=42).  $p$ -values ( $p_{\text{BH}}$ ) are computed against C0 using a bootstrap paired AUC difference test with Benjamini-Hochberg correction. For BACE,  $p$ -values reflect the statistical significance of the observed distributional shift, where performance degradation leads to near-chance ROC-AUC. Multi-seed stability results for C0 and C3 are provided in Table 2.

the zero-shot LLM predictions remain strictly below this traditional baseline, the relative differences between textual augmentation conditions reveal substantial semantic perturbation effects.

The LLM baseline (C0, SMILES only) yielded a ROC-AUC of 0.533 [0.471, 0.593]. Factual augmentation (C1) produced a marginal, non-significant increase to 0.541. Chemical priming (C2) yielded 0.613, while pure non-chemical gibberish (C2b) yielded 0.553.

Free hallucination (C3) improved ROC-AUC to 0.593 [0.519, 0.664]. Among category-specific conditions, Structural Phantom hallucination (C4a) yielded the highest observed ROC-AUC of 0.626 [0.548, 0.698], followed by Mechanism Fabrication (C4c, 0.611) and Property Inversion (C4b, 0.563).

**Bootstrap Stability.** To account for the single-split evaluation, we computed ranking stability across 1,000 bootstrap resamples (Figure 2). The superiority of SP hallucination (C4a) over the LLM baseline (C0) proved highly consistent, maintaining a positive margin in 97.6% of resamples. Free hallucination (C3) outperformed the baseline in 87.7% of resamples.

The random-permutation semantic control (C5) yielded 0.481 [0.408, 0.554], closely aligned with (and slightly below) the baseline C0 and substantially below C3. This indicates that the scientific register of hallucinated text alone does not account for the observed gains, and that disrupting semantic alignment effectively eliminates the predictive benefit of hallucinated descriptions.

#### 4.1.2 BACE (Enzymatic Task)

On the BACE test set ( $N = 152$ ), a contrasting pattern was observed. The baseline (C0) yielded a ROC-AUC of 0.674 [0.584, 0.756]. All hallucination-augmented conditions reduced performance

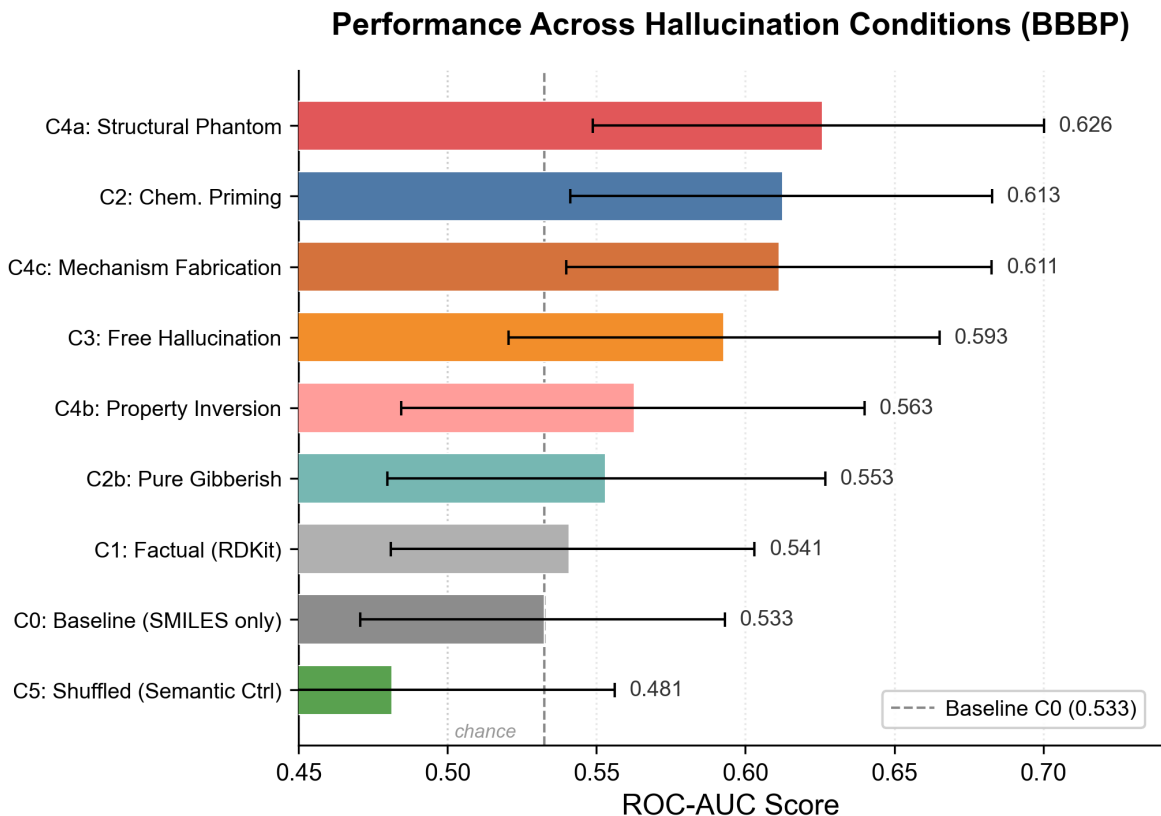


Figure 1: Task-dependent divergence in hallucination sensitivity across perturbation conditions, with structurally grounded (SP) hallucination associated with the largest discriminative improvement on BBBP.

relative to baseline. Free hallucination (C3) reduced ROC-AUC to 0.504 [0.409, 0.598], representing a substantial degradation toward near-chance behavior.

## 4.2 Distributional Compression on BACE

Under hallucination conditions on BACE, the distribution of predicted probabilities compressed from a bimodal pattern (reflecting the model’s baseline discriminative behavior) to a narrow unimodal distribution centered at  $p \approx 0.17$ , regardless of the molecule’s true label. This effect, which we descriptively term “distributional compression,” was observed across all hallucination conditions (C3, C4a–c) and suggests systematic signal destruction rather than random noise injection.

## 4.3 Distribution of Hallucination Types

Analysis of 356 naturally generated hallucinations across both datasets revealed a consistent distributional bias (Figure 4). Contextual Confabulation (CC) was the dominant category, accounting for approximately 80% of all generated descriptions ( $n = 163$  of 204 for BBBP). Structural Phantom (SP) was the rarest category at 2.4% ( $n = 5$  of 204 for BBBP). Despite this low natural frequency, forced SP hallucinations (C4a) exhibited the strongest predictive association on BBBP.

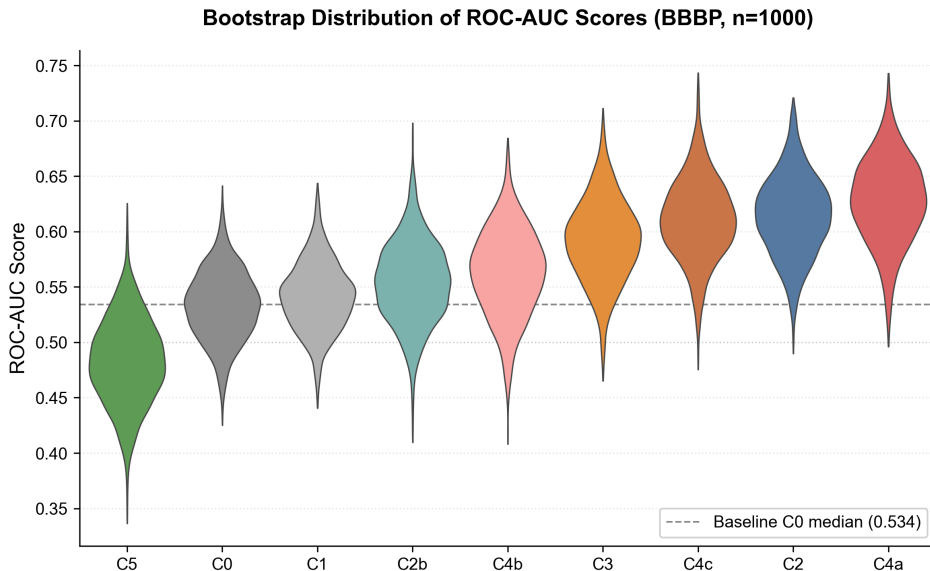


Figure 2: Bootstrapped ROC-AUC distributions demonstrate the statistical stability of Structural Phantom (C4a) hallucination gains relative to the SMILES baseline (C0) on the BBBP dataset.

#### 4.4 Robustness and Cross-Model Generalization

To assess the stability of the observed hallucination effects, we conducted multi-seed evaluations for the Llama-3.1-8B model ( $n = 3$  seeds) and cross-model validation using the larger Llama-3.3-70B model (Table 2).

On the BBBP task, the performance shift from free hallucination (C3) was consistent across seeds ( $\sigma = 0.006$ ), confirming that the gain is not a stochastic artifact. However, the larger Llama-3.3-70B model exhibited a different behavior: while its baseline (C0) performance was substantially higher (0.644), hallucination augmentation degraded its discriminative accuracy (0.549). This suggests that the observed directional shift may be specific to models with lower baseline chemical knowledge, whereas higher-capacity models are more sensitive to fabrication-induced noise.

On the BACE task, the "distributional compression" phenomenon was universally consistent. Both models and all seeds showed a significant degradation of ROC-AUC toward chance level ( $0.512 \pm 0.005$ ) when augmented with hallucinations, regardless of the baseline strength. This reinforces our finding that enzymatic property prediction is systematically fragile to semantic perturbations across LLM architectures.

Table 2: Robustness and generalization of the hallucination effect. Llama-8B values represent mean  $\pm$  standard deviation across three independent seeds (42, 123, 777). Llama-70B results are based on a single inference run (seed 42), precluding variance estimation.

| Task / Model                  | C0 (Baseline)     | C3 (Free Hallu)   | $\Delta$ AUC |
|-------------------------------|-------------------|-------------------|--------------|
| <i>BBBP (Physicochemical)</i> |                   |                   |              |
| Llama-8B ( $n = 3$ seeds)     | $0.502 \pm 0.021$ | $0.601 \pm 0.006$ | +0.099       |
| Llama-70B (Single seed)       | 0.644             | 0.549             | -0.095       |
| <i>BACE (Enzymatic)</i>       |                   |                   |              |
| Llama-8B ( $n = 3$ seeds)     | $0.644 \pm 0.021$ | $0.512 \pm 0.005$ | -0.132       |
| Llama-70B (Single seed)       | 0.564             | 0.506             | -0.058       |



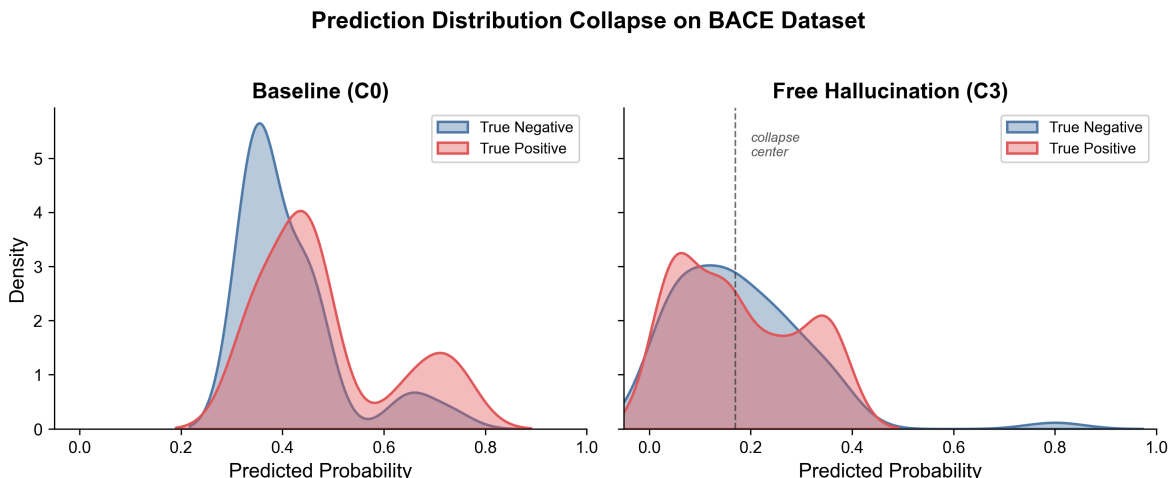


Figure 3: Hallucination augmentation induces pronounced distributional compression on the BACE task, shifting the baseline bimodal prediction distribution toward a non-discriminative narrow peak near  $p \approx 0.17$ .

## 5 Discussion

We organize this discussion by explicitly separating empirical observations from interpretive speculation, identifying alternative explanations, and establishing rigorous boundaries for our claims.

### 5.1 Distributional Bias in Hallucination Generation

**Observation.** Across both datasets, naturally generated hallucinations (C3) are overwhelmingly classified as Contextual Confabulation (CC,  $\sim 80\%$ ), with Structural Phantom (SP) accounting for only 1.3–2.4% of outputs. Despite this rarity, forced SP hallucinations (C4a) were associated with the largest observed ROC-AUC increase on BBBP, though we emphasize this is based on forced perturbations rather than naturally occurring frequencies.

**Interpretation.** We characterize this distributional asymmetry as a generative bias toward generic, scientifically plausible narratives over structurally specific confabulations. This labeling is purely descriptive and does not imply a causal generative mechanism within the LLM architecture.

**Alternative Explanations.** The low SP rate may reflect prompt design interactions rather than an intrinsic model bias; different prompt formulations or higher temperatures might elicit more structurally specific outputs. Additionally, our rule-based classifier may conservatively undercount SP instances if the generated text employs structural synonyms not captured by our heuristic keyword dictionary.

### 5.2 Task-Dependent Asymmetry

**Observation.** On BBBP, hallucination augmentation (C3, C4a–c) was associated with improved ROC-AUC relative to the SMILES-only baseline. The LLM baseline (C0, SMILES only) yielded a ROC-AUC of 0.533 [0.471, 0.593]. Factual augmentation (C1) produced a marginal, non-significant increase to 0.541. Chemical priming (C2) yielded 0.613, while pure non-chemical gibberish (C2b) yielded 0.553. Conversely, on BACE, all hallucination conditions degraded performance, with C3 reducing ROC-AUC from 0.674 to 0.504. The BACE prediction distribution exhibited pronounced compression, shifting from a bimodal pattern to a narrow unimodal distribution centered near  $p \approx 0.17$ .

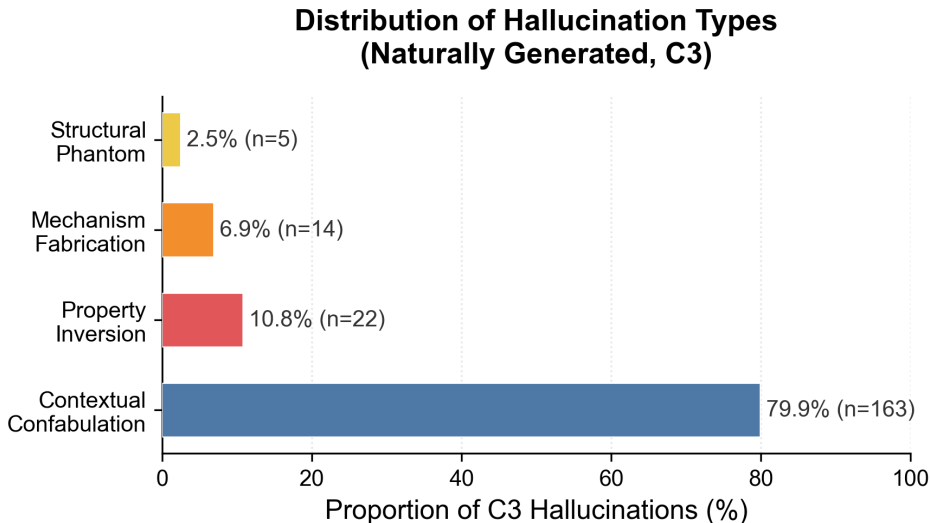


Figure 4: Free hallucinations exhibit a pronounced distributional bias toward generic Contextual Confabulation (CC), with structurally grounded perturbations (SP) occurring rarely despite their stronger association with predictive performance.

**Interpretation.** This task-dependent asymmetry suggests that hallucination utility is highly sensitive to the target domain. One plausible interpretation is that the LLM’s pre-training corpus contains more robust latent representations of general physicochemical properties (relevant to BBBP) than specific enzymatic mechanisms (BACE), leading to constructive textual perturbation in the former and destructive interference in the latter.

**Alternative Explanations.** We emphasize that these observed gains on BBBP do not demonstrate that hallucinations enhance genuine molecular reasoning. The improvements may partially reflect latent statistical correlations introduced by textual perturbation that happen to align with task-relevant physicochemical descriptors in the BBBP evaluation set. Notably, while textual perturbations induce directional shifts in raw prediction scores, none of the observed ROC-AUC improvements on the BBBP dataset reached statistical significance ( $p > 0.05$ ) after Benjamini-Hochberg correction. This suggests that while generic textual noise can induce consistent shifts in model confidence, the discriminative boost is modest and sensitive to the small sample size ( $N = 204$ ). The BACE degradation, by contrast, was statistically significant across most conditions, likely reflecting dataset-specific class imbalance interacting adversely with the hallucinated text. On BACE, the SP condition (C4a) was a borderline exception (ROC-AUC 0.666 vs. 0.674 baseline,  $p = 0.808$ ), suggesting that structural grounding might slightly mitigate the distributional compression observed in other hallucination types. We cannot distinguish among these mechanisms with the current experimental design.

### 5.3 Semantic Content vs. Stylistic Priming

**Observation.** The random-permutation control (C5) preserved scientific register while disrupting molecule-description semantic alignment. C5 produced no discriminative improvement over baseline (ROC-AUC 0.481 vs. 0.533 for C0 on BBBP), despite utilizing text that is linguistically indistinguishable from C3 hallucinations.

**Interpretation.** This dissociation is consistent with the hypothesis that the performance shifts observed under C3 are not attributable solely to generic scientific writing style or prompt length, but

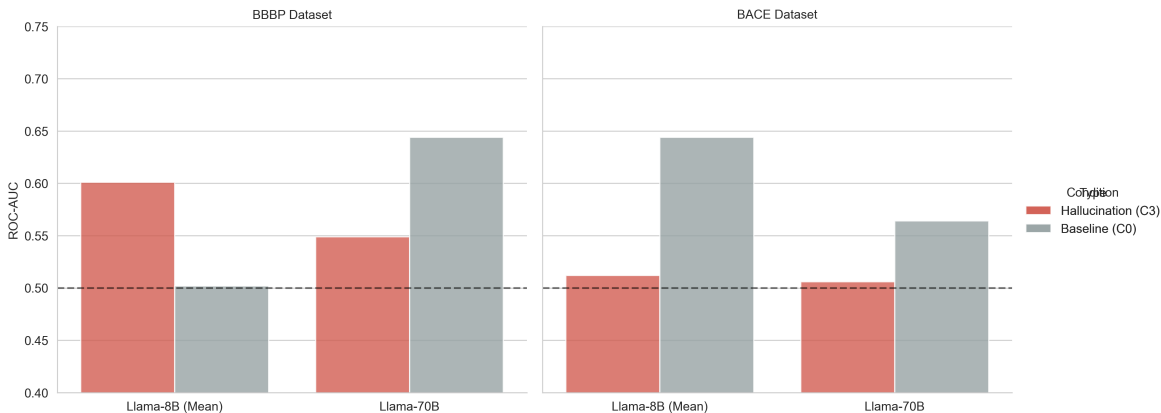


Figure 5: Architecture-dependent response to hallucination augmentation. Smaller models (8B) exhibit a consistent directional improvement on the BBBP task, whereas the larger model (70B) shows sensitivity to fabrication-induced noise, resulting in performance degradation.

depend on some degree of semantic interaction between the description and the target molecule.

**Boundary.** The C5 control uses a random permutation ( $seed = 42$ ). While this approach minimizes the risk of structural proximity bias common in simple circular shifts, future work should employ even more rigorous semantic isolation, such as adversarial embedding-based negative sampling or cross-model hallucinations, to further decouple register from relevance.

## 5.4 Architecture-Dependent Sensitivity

**Observation.** While the Llama-3.1-8B model exhibited a consistent performance boost under C3 for the BBBP task, the larger Llama-3.3-70B model showed a reversed effect, with hallucination augmentation degrading its strong baseline performance (0.644 to 0.549). On the BACE task, however, both model scales exhibited similar distributional compression and performance degradation under C3.

**Interpretation.** This divergence may suggest that sensitivity to semantic perturbations is non-linearly related to model capacity and baseline domain knowledge. One possible interpretation is that smaller models, possessing lower baseline chemical reasoning capacity, may rely more heavily on semantically enriched perturbations as a form of "noisy intuition," whereas larger models, having captured more robust latent chemical representations during pre-training, may be more sensitive to the noise introduced by fabrications. Interestingly, the larger model’s baseline on BACE (0.564) was lower than that of the 8B model (0.644), indicating that increased parameter scale does not uniformly translate to improved zero-shot performance across all chemical tasks. We offer these as tentative hypotheses, noting that further investigation into the interaction between model scale and perturbation sensitivity is required.

## 5.5 Scope and Limitations

To prevent overinterpretation, we explicitly bound our claims:

- We report observational associations within a controlled perturbation framework; we do not establish a causal mechanism for how hallucinations alter LLM predictions.
- While we include cross-model validation, this was limited to two conditions (C0, C3) and a single inference run for the Llama-3.3-70B model, precluding confidence interval estimation for the

larger architecture. The full nine-condition ablation across multiple seeds for the 70B model remains as future work.

- The observed utility of structurally grounded (SP) perturbations is strictly bounded to the BBBP dataset and the tested prompt templates. We do not claim this represents a generalized phenomenon across all molecular benchmarks.
- We do not propose hallucination augmentation as a practical, deployable method for molecular property prediction. The observed effects may be artifacts of the zero-shot evaluation setting rather than robust indicators of reasoning capacity. Note: Table 1 reports results for the primary single-run evaluation (seed=42).  $p$ -values ( $p_{BH}$ ) are computed against C0 using a bootstrap paired AUC difference test with Benjamini-Hochberg correction; hence, the baseline C0 comparison yield  $p = 1.000$ . For BACE,  $p$ -values reflect the statistical significance of the observed distributional shift, where performance degradation leads to near-chance ROC-AUC. Multi-seed stability results for C0 and C3 are provided in Table 2.
- The proposed categorization scheme is operational, not exhaustive. The CC category is a residual class, and finer-grained semantic ontologies are required for comprehensive analysis.
- We note that traditional machine learning approaches, such as the Random Forest baseline reported in Section 4, are likely to achieve substantially higher predictive performance than the zero-shot LLM settings evaluated here.

Ultimately, these findings highlight that LLM-based molecular prediction pipelines can exhibit non-trivial sensitivity to semantically structured perturbations, suggesting that seemingly benign textual augmentations may introduce systematic biases in downstream predictions. Understanding this sensitivity is critical for interpreting LLM-assisted predictions, particularly in settings where textual inputs are used as auxiliary signals.

## 6 Limitations

This study has several important limitations that constrain the generalizability and interpretability of its findings.

**Model scope.** Primary experiments were conducted with Llama-3.1-8B-Instant. While cross-model validation was performed with Llama-3.3-70B, this was limited to two conditions (C0, C3) and a single inference run, precluding a full nine-condition multi-seed ablation for the larger architecture. The observed effects—including the task-dependent performance asymmetry—may be sensitive to model family, parameter count, or inference backend beyond the two evaluated scales.

**Single-split evaluation.** Results are based on a single scaffold-split test set for each dataset (BBBP  $N = 204$ , BACE  $N = 152$ ) without cross-validation. This makes the reported ROC-AUC values sensitive to the specific test partition and limits confidence in the stability of the observed rankings.

**Limited sample size.** With 204 and 152 molecules respectively, this study is underpowered for detecting small but potentially meaningful effect size differences between conditions. The SP category comprised only 5 naturally occurring instances in BBBP, making any distributional claims about SP highly tentative.

**C2 control confound.** The chemical priming condition (C2) contains domain-relevant vocabulary (“structural”, “molecular”, “physicochemical”), making it a partial rather than pure stylistic control. While C2b addresses this with non-chemical vocabulary, the C2 results should be interpreted with this confound in mind.

**Residual taxonomy category.** The Contextual Confabulation (CC) category functions as a catch-all residual class for descriptions that do not trigger SP, PI, or MF detection rules. This category likely encompasses heterogeneous subtypes with different functional properties, and its homogeneous treatment may obscure important variation.

**Data leakage risk.** The Llama-3.1 pre-training corpus may include MoleculeNet datasets or related molecular property data. We cannot rule out that the model’s predictions reflect memorized associations rather than genuine zero-shot inference. This risk applies to all conditions but may differentially affect conditions where hallucinated text inadvertently retrieves memorized information.

**API non-determinism.** Despite setting temperature = 0.0 and seed = 42, Groq’s API documentation notes that seed-based reproducibility is provided on a best-effort basis. Exact numerical reproducibility across runs is therefore not guaranteed.

**Dataset scope.** Only two MoleculeNet benchmarks were evaluated, representing physicochemical (BBBP) and enzymatic (BACE) prediction tasks. Generalization to other task types (e.g., toxicity, solubility, protein binding) remains untested.

**No fine-tuning baseline.** We evaluate only zero-shot LLM inference. Comparison with fine-tuned models or traditional ML baselines (e.g., random forests on molecular fingerprints) would provide important context for the practical significance of the observed effects.

## 7 Conclusion

The primary contribution of this work is a controlled semantic perturbation framework that enables systematic isolation of hallucination effects in LLM-based molecular prediction. This exploratory study provides empirical evidence that the type of hallucination—not merely its presence—is associated with differential outcomes in LLM-augmented molecular property prediction. Through a controlled nine-condition ablation using a four-category categorization scheme, we observe that Structural Phantom (SP) hallucinations exhibit the strongest predictive association on the BBBP physicochemical task (ROC-AUC 0.626 vs. 0.533 baseline), despite being naturally underproduced by the model (approximately 2.4% of free hallucinations). Conversely, on the BACE enzymatic task, hallucination augmentation consistently degraded performance, suggesting that augmentation utility is sensitive to the alignment between the perturbation and the task’s knowledge requirements.

Our random-permutation semantic control (C5) provides evidence that the observed gains on BBBP are not attributable solely to scientific writing style, as disrupting molecule-description alignment eliminates discriminative improvement while preserving linguistic register.

These findings should be interpreted cautiously given the limited scope of this pilot study: two LLM scales, two datasets, and single-split evaluation. We do not propose hallucination augmentation as a practical technique, nor do we claim to have identified the causal mechanism underlying the observed effects. Rather, this work aims to provide a structured empirical characterization that can guide future investigation into when and why semantic perturbations affect LLM-based molecular reasoning.

**Future directions.** Key extensions include: (1) investigation of the distributional compression phenomenon on additional enzymatic datasets, (2) expansion of multi-model validation across diverse LLM architectures and scales, (3) larger-scale evaluation with cross-validation, and (4) comparison with fine-tuned and traditional ML baselines to contextualize the practical significance of the observed effects.

## Code Availability

The complete codebase, datasets, and high-resolution figures are available at: <https://github.com/apiprdt/AI-Hallucinations>

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## Data Availability

The raw model responses, processed datasets, and source code used for this study are available at: <https://github.com/apiprdt/AI-Hallucinations>. A static version of the evaluation datasets is provided within the repository to ensure exact reproducibility.

## Ethics Statement

No ethical approval was required for this computational study as it did not involve human subjects, animal testing, or sensitive personal data.

## Competing Interests

The author declares no competing interests.

## Grant Information

No specific funding was received for this work.

## A Appendix

### A.1 Prompt Templates

All prediction tasks used the following zero-shot prompt structure with temperature = 0.0, max\_tokens = 10, and seed = 42:

Given this molecule (SMILES: {smiles})  
[Additional information: {augmentation}]

On a scale of 0 to 1, what is the probability  
that this molecule can {task\_description}?  
Respond with ONLY a number between 0 and 1.  
Nothing else.

The augmentation line is omitted for C0. Task descriptions: “penetrate the blood-brain barrier” (BBBP), “inhibit the BACE-1 enzyme” (BACE).

Hallucination generation used temperature = 0.9, max\_tokens = 150, with the following type-specific prompts:

**C3 (Free Hallucination):** *“You are a speculative chemist. Given SMILES: {smiles}. Generate a creative, scientifically-sounding description of this molecule’s potential properties and biological mechanisms. Be creative and speculative. Write 2–3 sentences only.”*

**C4a (Structural Phantom):** *“Given SMILES: {smiles}. Describe the structural features, functional groups, and atomic composition. Be specific and speculative.”*

**C4b (Property Inversion):** *“Given SMILES: {smiles}. Describe its solubility, lipophilicity, and membrane permeability with specific values.”*

**C4c (Mechanism Fabrication):** *“Given SMILES: {smiles}. Hypothesize biological targets and cellular mechanisms. Be specific about protein targets.”*

### A.2 Control Condition Templates

**C2 (Chemical Priming):** Template-generated text using scientific vocabulary: *“This compound exhibits {adj} structural coherence with {noun} interaction domains, demonstrating {adj} physicochemical characteristics under {noun} conditions.”*

Adjective pool: moderate, intermediate, variable, standard, conventional, typical. Noun pool: hypothetical, theoretical, computational, structural, molecular, chemical.

**Note:** This template contains chemical vocabulary (“structural”, “molecular”, “physicochemical”), making C2 a *partial* stylistic control. See C2b for the stricter control.

**C2b (Pure Gibberish):** Template-generated text using exclusively non-chemical vocabulary: *“This entity exhibits {adj} portfolio coherence with {noun} governance domains, demonstrating {adj} bureaucratic characteristics under {noun} conditions.”*

Adjective pool: variable, standard, conventional, typical, moderate, intermediate. Noun pool: portfolio, transactional, governance, bureaucratic, administrative, regulatory.

Vocabulary audit confirmed 0% overlap with a curated chemical/biological term list.



### A.3 Taxonomy Classifier Pseudocode

1. **SP Detection:** For each element in {F, Cl, Br, I, P, S, N}, check if associated keywords appear in the generated text. Cross-reference with `mol.GetAtoms()` to verify element absence. If a keyword matches an absent element, classify as SP.
2. **PI Detection:** Compute MolLogP via RDKit. If  $\text{LogP} > 3$  and text contains “water soluble”, “hydrophilic”, or “highly soluble”, classify as PI. If  $\text{LogP} < 1$  and text contains “lipophilic”, “hydrophobic”, or “membrane permeable”, classify as PI.
3. **MF Detection:** Keyword matching for 9 biological targets: ACE2, HER2, EGFR, VEGF, p53, BCL-2, COX-2, dopamine, serotonin.
4. **CC (Default):** If none of the above conditions trigger, classify as Contextual Confabulation.

### A.4 Representative Hallucination Examples

- **SP Example (SMILES: c1ccccc1, Benzene):** “Contains a reactive chlorine atom that enhances metabolic stability.” (Benzene contains no chlorine atoms.)
- **PI Example (SMILES: high-LogP compound):** “This highly water-soluble molecule readily dissolves in aqueous media.” (Contradicts computed  $\text{LogP} > 3$ .)
- **MF Example (Diazepam):** “Acts as a potent inhibitor of the VEGF protein kinase receptor.” (Diazepam is a GABA receptor modulator, not a VEGF inhibitor.)
- **CC Example:** “This molecule shows promising drug-like properties and may have therapeutic applications in treating various diseases.” (Generic, non-falsifiable.)

### A.5 API Failure Rates

Across the BBBP experiment ( $204 \text{ molecules} \times 9 \text{ conditions} = 1,836 \text{ predictions}$ ), API failures (rate limits, parsing errors) that defaulted to  $p = 0.5$  accounted for less than 2% of total predictions. The exact count is logged in the experiment checkpoint files.

### A.6 Implementation Environment

Experiments were executed on a local system (Windows 11) using the Groq API for LLM inference. Total runtime for the pilot study was approximately 4.5 hours. Library versions: `rdkit-pypi` v2023.9.1, `deepchem` v2.7.1, `groq` Python SDK, `scipy` v1.11, `scikit-learn` v1.3.